Project Schedule
and Critical Path Analysis

Project Name: AI Task Breakdown
Course: ENSF 400
Section: L01
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Group #: 14

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1. Activity Definition and Sequencing

WBS code	Work Package Name	Activities
1.1.1	Ideation and Concept Development	Brainstorm project ideas, Decide target audience, Finalize project concept
1.1.2	Project Kickoff Presentation	Draft initial presentation content, Design and format presentation slides, Deliver initial presentation
2.1.1	SRS Document	Gather and document functional requirements, Document non-functional requirements, Review and finalize SRS document
2.2.1	System Architecture Diagram	Design system architecture diagram, Validate architecture against requirements,
2.2.2	Database Schema Diagram	Identify and map database entities
2.3.1	Wireframes	Design wireframes for Dashboard screen, Task Viewer Screen and Login Screen,
3.1.1	Project Description Input	Build project description text area component, Implement input validation logic
3.1.2	Task Viewer	Implement task list and Kanban UI
3.1.3	Conversational Chat Interface	Build chat input, Implement chat error handling
3.1.4	Export Tasks	Implement task export for pdf or xlsx
3.2.1	User Authentication	Implement user registration
3.2.2	LLM API Integration	Integrate LLM via API
3.2.3	Persistent Tasks	Integrate database via API
4.1.1	Unit Testing	Write unit tests for Auth module, Write unit tests for LLM parsing
4.1.2	Integration Testing	End-to-end integration testing
4.1.3	System Testing	Execute a full system test, Document system test results
4.2.1	User Acceptance Testing	Conduct UAT sessions, Collect and document UAT sign-offs
4.2.2	Defect Tracking	Defect Tracking/Fixing
5.0.1	Frontend Deployment	Frontend Vercel Deployment, Validate frontend deployment
5.0.2	Backend Deployment	Backend deployment, Validade backend deployment
5.0.3	Database Deployment	Configure and deploy database to Supabase, Validate database deployment
6.0.1	Collect Feedback and Document Lessons Learned	Compile project feedback
6.0.2	Final Project Report	Draft final project report, Review and Submit final project report

1.1. Activity List

Activity ID	Activity Name	Related WBS ID	Description	Estimated Duration
A1.1.1a	Brainstorm project ideas	1.1.1	Conduct team brainstorming session to generate and evaluate potential project concepts	1 hr
A1.1.1b	Define target audience	1.1.1	Identify and document the intended users and their needs	1 hr
A1.1.1c	Finalize project concept	1.1.1	Select and document the project direction and scope	0.5 hr
A1.1.2a	Draft initial presentation content	1.1.2	Write and organize content covering objectives, scope, timeline, and roles	1 hr
A1.1.2b	Design and format presentation slides	1.1.2	Build the PowerPoint with visuals, layout, and speaker notes	1 hr
A1.1.2c	Deliver initial presentation	1.1.2	Present the finalized presentation to stakeholders	0.2 hr
A2.1.1a	Gather and document functional requirements	2.1.1	Interview team members and define all functional requirements (FR-1 through FR-6)	2 hr
A2.1.1b	Document non-functional requirements	2.1.1	Define performance, reliability, usability, and security requirements.	2 hr
A2.1.1c	Review and finalize SRS document	2.1.1	Proofread, validate, and submit the completed SRS document	1 hr
A2.2.1a	Design system architecture diagram	2.2.1	Map the full technical blueprint including UI, backend, and LLM API data flow	3 hr
A2.2.1b	Validate architecture against requirements	2.2.1	Review diagram against LLM latency and SRS constraints.	1 hr
A2.2.2a	Identify and map database entities	2.2.2	Define all entities for users, projects, and tasks in Supabase.	4 hr
A2.3.1a	Design wireframes for Login Screen	2.3.1	Create a wireframe for the user authentication screen.	2 hr
A2.3.1b	Design wireframes for Dashboard screen	2.3.1	Create a wireframe for the main project dashboard.	2 hr
A2.3.1c	Design wireframes for Task Viewer Screen	2.3.1	Create a wireframe for the task list and Kanban view.	2 hr
A2.3.1d	Review all	2.3.1	Verify and check that all screens are covered, compatible with mobile and	0.5 hr

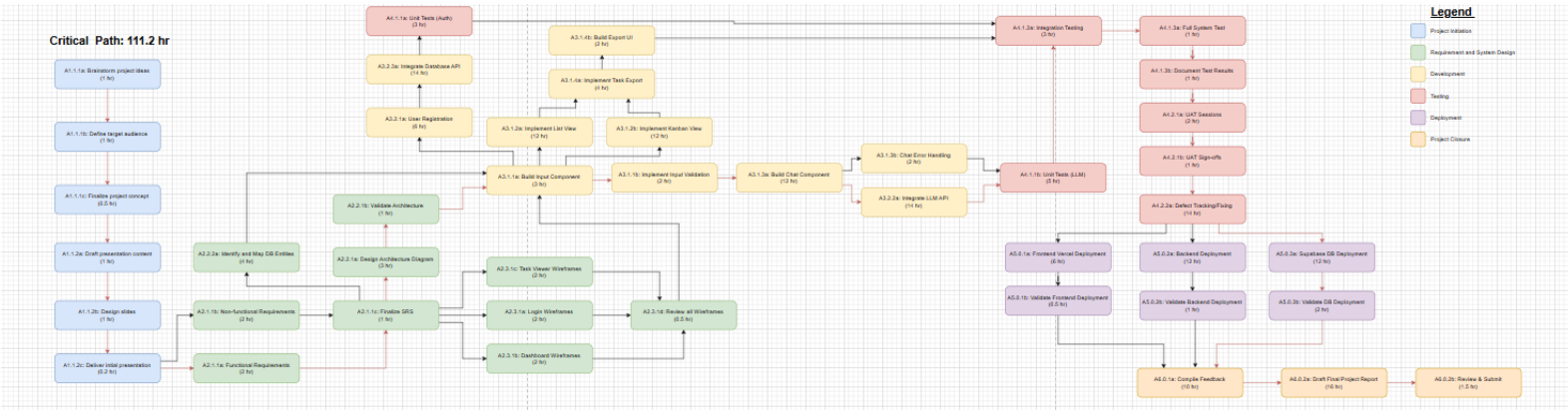
	wireframes		desktop, and are visually consistent.	
A3.1.1a	Build project description text area component	3.1.1	Develop the frontend input field for user project prompt submission	3 hr
A3.1.1b	Implement input validation logic	3.1.1	Add empty-input checks and loading state feedback to the submission flow	2hr
A3.1.2a	Implement task list view UI	3.1.2	Build the structured list format for displaying generated tasks	12 hr
A3.1.2b	Implement Kanban board view UI	3.1.2	Build the card-based Kanban view and toggle between list and board modes	12 hr
A3.1.3a	Build conversational chat input component	3.1.3	Create the natural language input field for modifying tasks via LLM commands	12 hr
A3.1.3b	Implement chat error handling	3.1.3	Display error messages when conversational LLM updates fail	2 hr
A3.1.4a	Implement task export	3.1.4	Develop backend logic that formats and exports task as a PDF, CSV, and JSON	4 hr
A3.1.4b	Build task export UI controls	3.1.4	Add UI buttons and selection options for different types of file exports	2 hr
A3.2.1a	Implement user registration	3.2.1	Build email, password sign-up logic and form using Supabase DB.	6 hr
A3.2.2a	Integrate LLM via API	3.2.2	Integrate the LLM into the backend of the application	14 hr
A3.2.3a	Integrate database via API	3.2.3	Integrate the database into the backend of the application for data persistence of users	14 hr
A4.1.1a	Write unit tests for Auth module	4.1.1	Develop and run unit tests covering logins, signups and session logic.	3 hr
A4.1.1b	Write unit tests for LLM parsing	4.1.1	Develop and run unit tests for task extraction and LLM responses	5 hr
A4.1.2a	End-to-end integration testing	4.1.2	Implement basic integration tests and validate the results through both automated tests and manual verification	3 hr
A4.1.3a	Execute a full system test	4.1.3	Run end to end tests that cover logins, task generation, modification, and exporting.	1 hr
A4.1.3b	Document system test results	4.1.3	Record all the pass/fail results and flag any issues for the defect log	1 hr
A4.2.1a	Conduct UAT sessions	4.2.1	Gather test participants and have them go through the whole task breakdown workflow with no prior training.	2 hr
A4.2.1b	Collect and document UAT sign-offs	4.2.1	Gather feedback from participants and obtain UAT approval	1 hr
A4.2.2a	Defect Tracking/Fixing	4.2.2	Identify, log and fix bugs found during testing	14 hr
A5.0.1a	Frontend Vercel Deployment	5.0.1	Set up Vercel pipeline and deploy client-side code to production HTTPS URL	6 hr

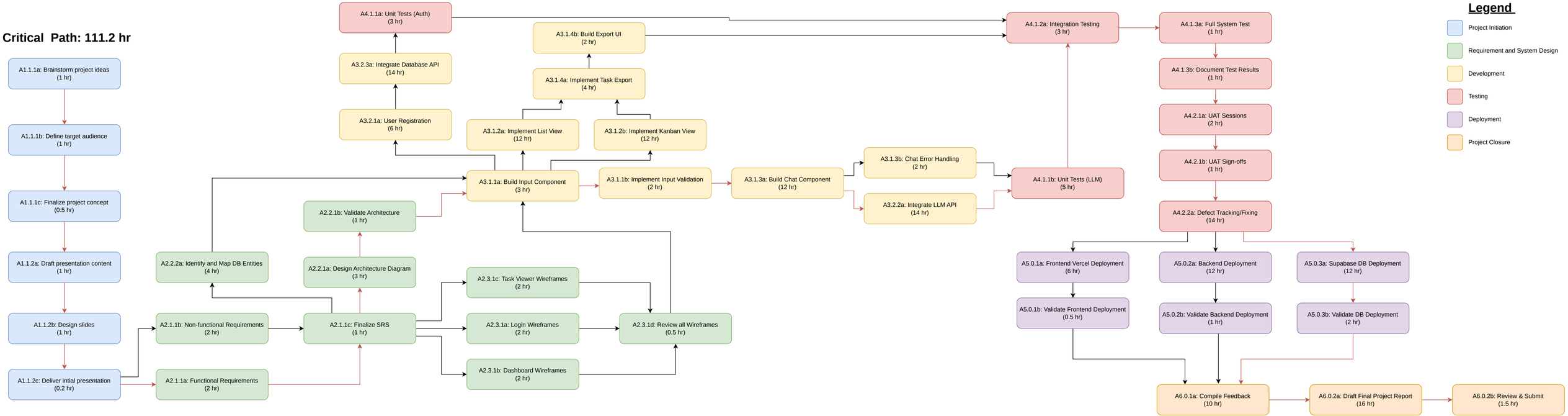
A5.0.1b	Validate frontend deployment	5.0.1	Test live URL for page reload speed under 5 seconds for both desktop and mobile.	0.5 hr
A5.0.2a	Backend deployment	5.0.2	Host the backend logic with live API endpoints	12 hr
A5.0.2b	Validade backend deployment	5.0.2	Test API endpoints of live deployed backend	1 hr
A5.0.3a	Supabase database deployment	5.0.3	Finalize and push the live Supabase production instance for persisting user data	12 hr
A5.0.3b	Validate database deployment	5.0.3	Confirm task changes persist for logged in users after page reload.	2 hr
A6.0.1a	Compile project feedback	6.0.1	Document project successes, failures, and lessons learned	10 hr
A6.0.2a	Draft final project report	6.0.2	Summarize project execution and final outcome into a report	16 hr
A6.0.2b	Review and Submit final project report	6.0.2	Proofread along with the rubric's requirements by March 23rd, 2026	1.5 hr

2. Network Diagram and Critical Path

Critical Path: 111.2 hrs

See the below image for the Network Diagram and the appendix for clearer/larger diagram:



Appendix

Critical Path: 111.2 hr